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Abstract

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1 Introduction

We ask three questions:

RQ1 *Do model behaviors reflect inferences about implicit causal relationships within sentences?*
RQ2 *Do LMs form internal representations of implicit causal relationships?*
RQ3 *If **RQ2** is true, do representations of implicit causality generalize to naturalistic settings with potential real-world implications?*

Our contribution is...

2 Background

3 Experiments

3.1 Measuring Implicit Causal Inference through Model Behaviors

Models. We test Gemma 2 2B/9B [Gemma Team, 2024], Qwen3 4B/8B [Qwen Team, 2025], Llama 3.2 3B [Meta AI, 2024], and Llama 3 8B [Dubey et al., 2024] base checkpoints.

Dataset.

Methods.

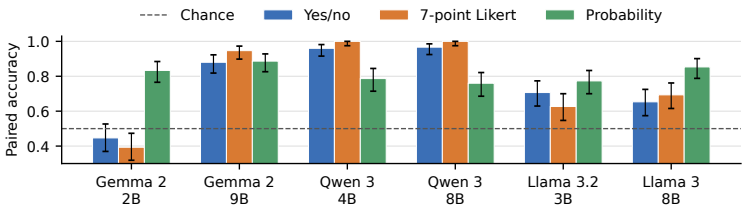


Figure 1: Paired accuracy on 150 pairs of implicit causal inference items with 95% CIs, **Models demonstrate implicit causal inference consistently through probability.** Not all models can faithfully report their implicit causal inference through Yes/No and 7-Point Likert rating. The performance of metalinguistic causal inference increases with model scale.

Results.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

3.2 Identifying Internal Representations of Implicit Causality Inference

We next ask whether model activations distinguish causal relationships from other relations. If a representation is specific to causality, a causal direction should separate causal pairs well but transfer less reliably to temporal or plausibility pairs.

Datasets.

CAA directions. Contrastive Activation Addition (CAA) represents a relation with a direction in a model's activation space [Rimsky et al., 2024]. At each tested layer, we subtract the final-token activation of every negative item from that of its matched positive item, then average these differences over the training pairs. An evaluation item receives a scalar score by projection onto this direction of averaged difference; paired accuracy is the proportion of evaluation pairs for which the positive item scores higher.

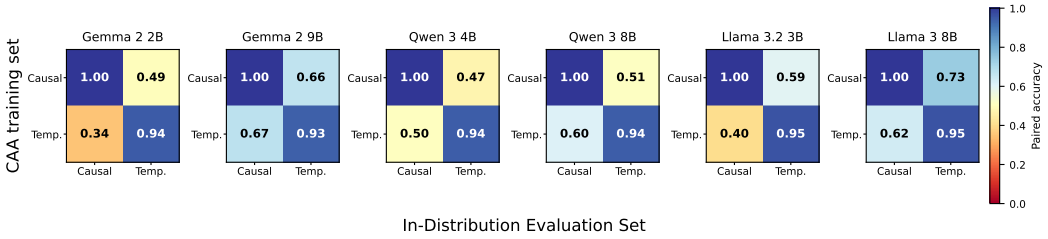


Figure 2: Cross-transfer paired accuracy between causal and temporal CAA directions. Rows show the CAA training set and columns show the in-distribution evaluation set. **Causal and temporal CAA directions are largely relation-selective.** Both directions perform best on their own relation and transfer less consistently to the other relation.

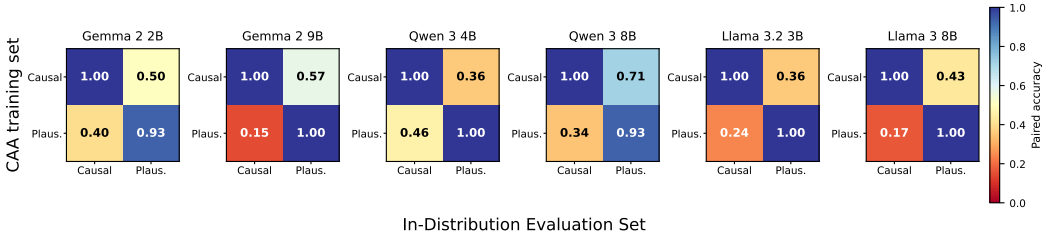


Figure 3: Cross-transfer paired accuracy between causal and plausibility CAA directions. Rows show the CAA training set and columns show the in-distribution evaluation set. **Implicit causality is represented distinctly from event plausibility.** Both directions perform best on their own relation; results on the 14-pair plausibility evaluation set are exploratory.

Results.

3.3 Testing the Generalization to Within-Sentence Implicit Causal Inference

We now ask whether the learned directions remain informative on unseen causal constructions within a single sentence (eliciture).

Training and evaluation data.

Analysis.

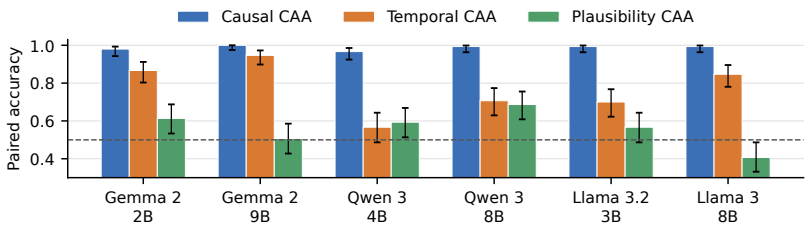


Figure 4: Paired accuracy of causal, temporal, and plausibility CAA directions on 150 unseen within-sentence causal inference pairs with 95% CIs. **Causal CAA directions generalize consistently to within-sentence implicit causal inference.** Temporal and plausibility CAA directions transfer less consistently across models. The dashed line denotes chance.

Results.

3.4 Study 4: Generalization to Naturalistic Scenarios

Dataset.

Table 1: Mean paired differences in causal CAA projection from the White-male condition across 65 matched scenarios, reported ± the bootstrap 95% CI half-width. **Demographic descriptions are associated with differences in causal CAA projections.** Positive values indicate greater projection than the White-male condition. Bolded values have CIs that do not include zero.

	Gemma 2 2B	Gemma 2 9B	Qwen3 4B	Qwen3 8B	Llama 3.2 3B	Llama 3 8B
Minority male − White male	**.41 ± .32**	**1.9 ± .40**	.073 ± .093	**.29 ± .17**	.025 ± .029	**.030 ± .020**
Minority female − White male	**.57 ± .32**	**2.3 ± .41**	**.15 ± .10**	**.39 ± .15**	**.069 ± .033**	**.060 ± .021**

Results.

4 Discussion

References

Abhimanyu Dubey et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024. URL `https://arxiv.org/abs/2407.21783`.

Gemma Team. Gemma 2: Improving open language models at a practical size. *arXiv preprint arXiv:2408.00118*, 2024. URL `https://arxiv.org/abs/2408.00118`.

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Qwen Team. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025. URL `https://arxiv.org/abs/2505.09388`.

Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering Llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15504–15522. Association for Computational Linguistics, 2024.

A Raw CAA Projection Distributions

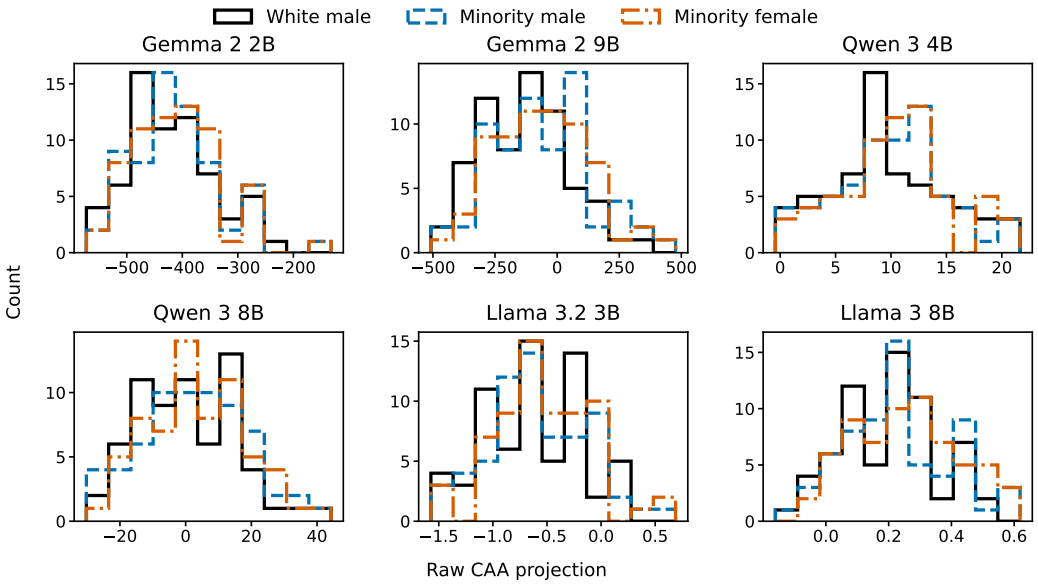


Figure 5: Raw causal CAA projection distributions for three matched demographic conditions across 65 scenarios per model. **The three conditions have substantially overlapping projection distributions in every model.** Colors and line styles distinguish the White-male, minority-male, and minority-female conditions.